21ECP302L – MINOR PROJECT

FINAL REVIEW

Mobile App-Controlled Reaction OLED System for Hand Rehabilitation and Motor Training

By

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PRESENTATION OUTLINE

S.No	Title
1	Abstract
2	Motivation
3	Literature Survey
4	Objectives
5	Innovation & Novelty
5	Conclusion
6	References

ABSTRACT

With the growing need for effective and technology-assisted rehabilitation methods, this project presents the design and development of a smart reaction training system for hand rehabilitation and cognitive–motor assessment. The proposed system uses visual stimulus-based exercises to improve hand coordination, reflex response, and neuromuscular performance. A microcontroller-based hardware setup is used to detect user interaction during training tasks. A TFT display is incorporated to project important information such as score, timing feedback, and task status in real time, enhancing user understanding and engagement. Instead of a dedicated mobile application, a web-based interface is used to control the system and calculate performance parameters including reaction time and task completion. This enables flexible access through smartphones and simplifies data monitoring. The developed system is portable, cost-effective, and suitable for both clinical and home-based rehabilitation environments, supporting structured therapy and continuous progress evaluation

MOTIVATION

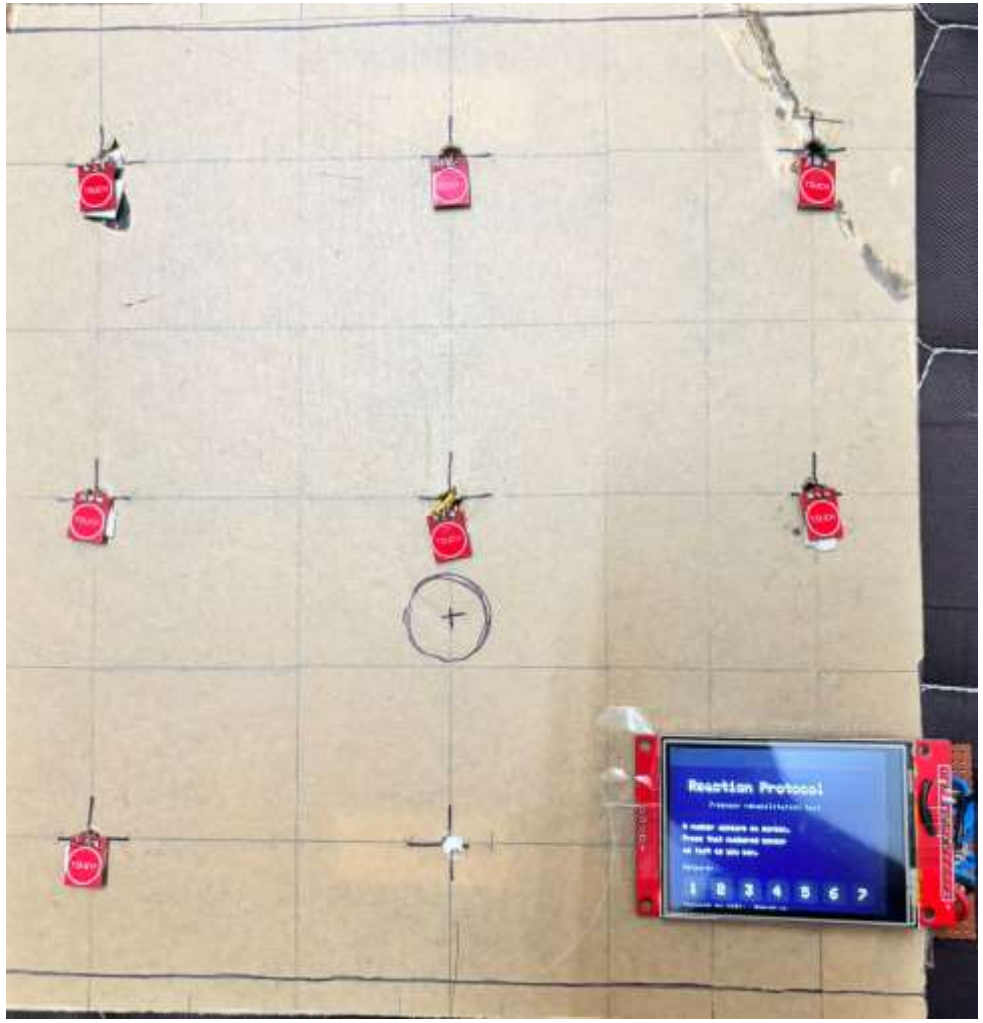
Hand impairments caused by stroke, injury, or post-surgical conditions significantly affect an individual's independence and quality of life. Traditional rehabilitation methods often require continuous supervision, repetitive exercises, and limited objective performance tracking, which can reduce patient engagement and slow functional recovery. The increasing availability of smart devices and low-cost embedded technologies creates an opportunity to develop interactive and data-driven rehabilitation solutions. This project is motivated by the need to provide an accessible, engaging, and measurable hand therapy system that encourages consistent practice and improves cognitive–motor coordination. By integrating app-based control with visual cue–driven reaction training, the proposed system aims to enhance patient motivation, enable personalized rehabilitation protocols, and support both clinical and home-based therapy. Ultimately, the motivation is to contribute to improved rehabilitation outcomes, faster recovery, and better monitoring of patient progress using simple and affordable technology solutions.

LITERATURE SURVEY

Sl.no	Title, journal name	Author	Year	Content
1	Smart Rehabilitation Systems Using IoT and Mobile Applications	Zhang et al	2024	This study presents an IoT-based rehabilitation platform that uses wearable sensors and mobile applications to monitor patient hand movements. The system enables real-time data collection, remote monitoring, and adaptive therapy exercises, improving patient engagement and rehabilitation outcomes.
2	Reaction Time Assessment Using Visual Cue-Based Training Devices	Kumar and Singh	2023	The research focuses on visual stimulus-based training systems designed to enhance cognitive–motor coordination. The proposed device measures reaction time and movement accuracy, showing significant improvement in neuromuscular rehabilitation performance.

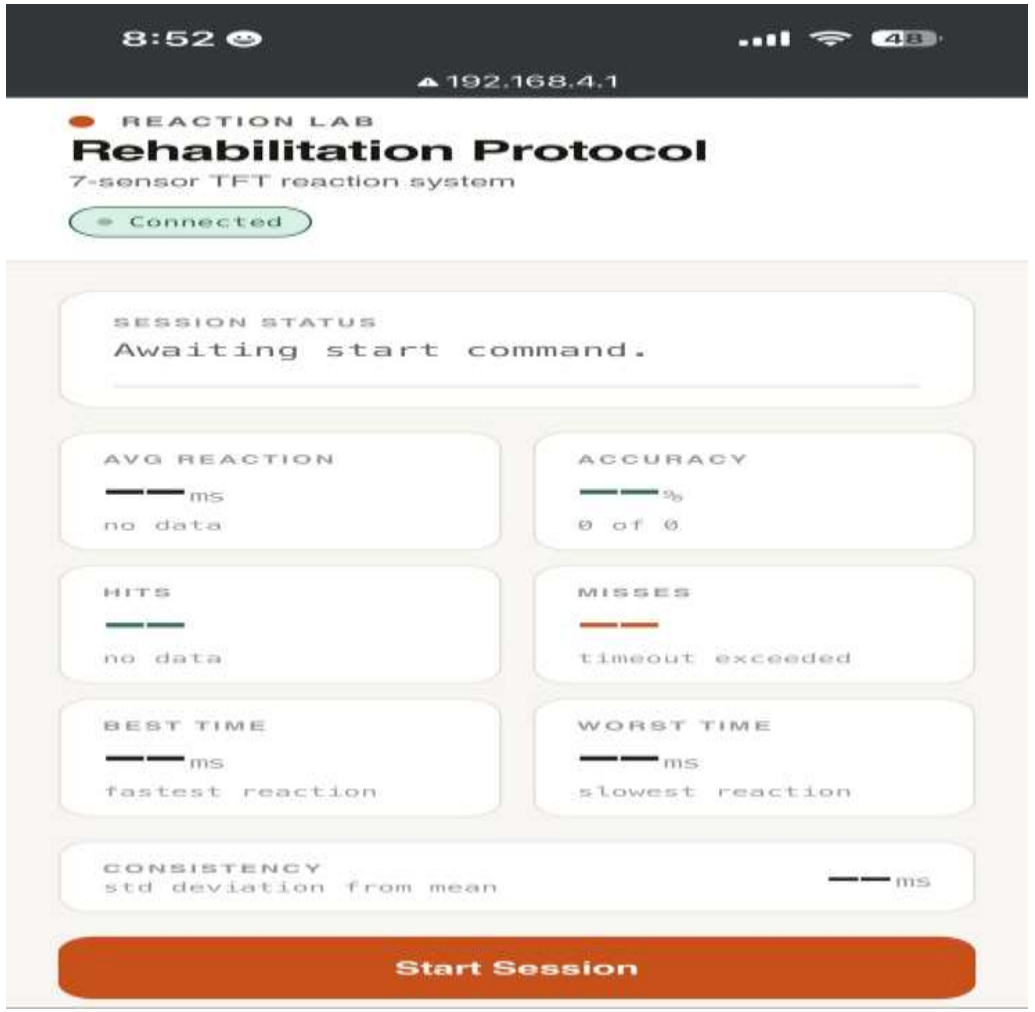
Sl.no	Title	Author	Year	Content
3	Mobile Health Applications for Upper Limb Rehabilitation	Lee et al	2023	This paper reviews mobile health solutions that provide structured therapy exercises and progress tracking. The results indicate that app-based rehabilitation increases therapy adherence and enables personalized treatment plans for stroke and injury patients.
4	Sensor-Based Smart Therapy Devices for Functional Recovery	Fernandez et al	2022	The study introduces a smart therapy device integrating touch sensors and wireless communication for monitoring hand interaction tasks. Experimental results demonstrate improved motor skill recovery.
5	Gamified Rehabilitation Systems for Cognitive and Motor Training	Park and Kim.	2022	This research explores gamification techniques in rehabilitation devices using visual cues and performance feedback. The findings highlight increased patient motivation, reduced therapy fatigue, and enhanced cognitive–motor response during repetitive training sessions

PROPOSED SYSTEM MODEL





REACTION SYSTEM - SOFTWARE



OBJECTIVES

•To design and develop a smart reaction training system for improving hand rehabilitation and cognitive–motor coordination.

•To implement visual stimulus-based training tasks that encourage timely hand interaction and repetitive movement practice.

•To integrate a microcontroller-based hardware system for detecting user responses and processing rehabilitation data.

•To incorporate a TFT display for projecting real-time performance information such as score, timing feedback, and task status.

•To develop a web-based interface that enables system control, performance calculation, and monitoring through smartphones.

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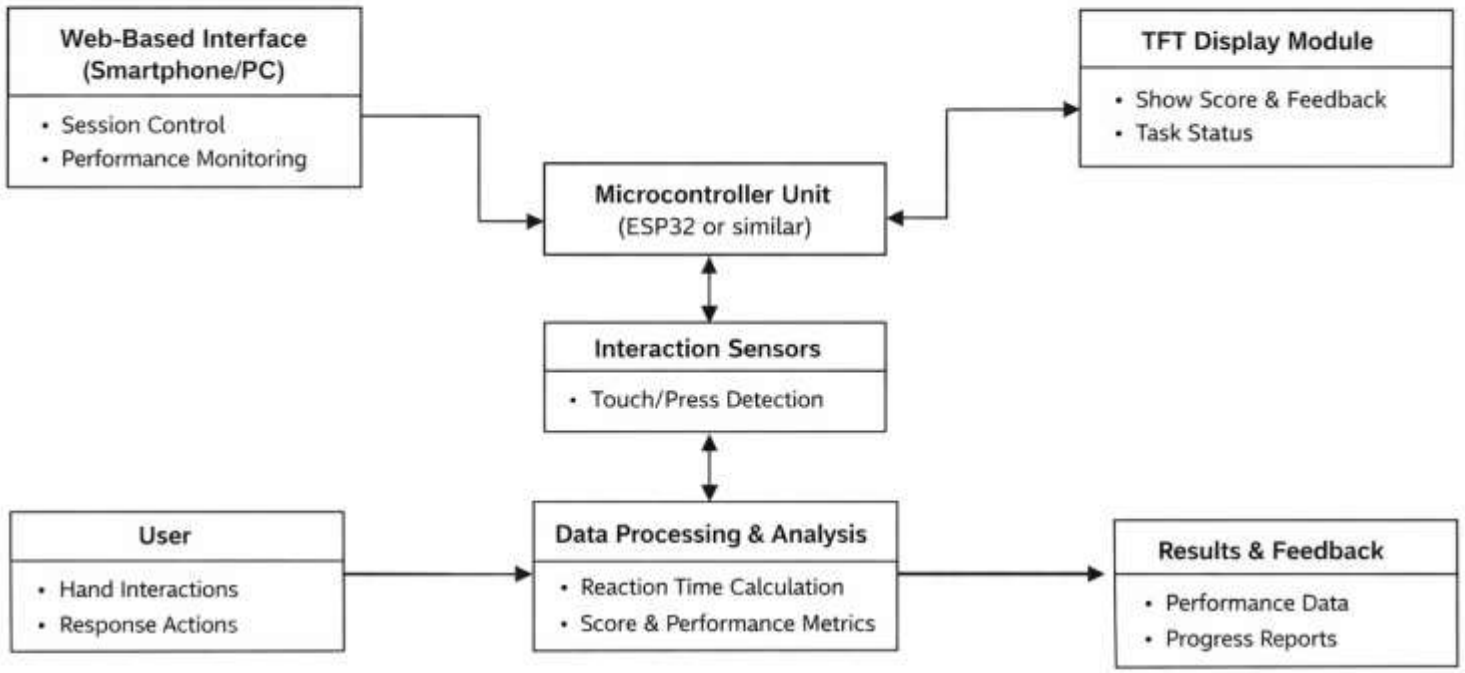
•To measure key rehabilitation parameters including reaction time, response accuracy, and task completion rate.

•To provide a portable and cost-effective rehabilitation solution suitable for both clinical therapy and home-based training environments

INNOVATION & NOVELTY

The proposed smart reaction training system introduces an innovative approach to hand rehabilitation by combining visual stimulus-based exercises with real-time performance monitoring using modern embedded and web technologies. Unlike conventional rehabilitation methods that rely on repetitive manual exercises and limited feedback, the developed system provides an interactive training environment that enhances user engagement and motivation. A key novelty of the project is the integration of a TFT display to project real-time information such as score, timing feedback, and task status, enabling users to clearly understand their performance during rehabilitation sessions. Additionally, the use of a web-based interface for system control and performance calculation eliminates the need for a dedicated mobile application and allows flexible access through smartphones or other smart devices. The system also enables objective assessment of rehabilitation parameters such as reaction time and task completion. Its portable, low-cost, and user-friendly design makes it suitable for both clinical and home-based therapy, contributing to the development of accessible and technology-driven rehabilitation solutions

PROPOSED SYSTEM MODEL



Smart Reaction Training System for Hand Rehabilitation

REFERENCES

[1] Y. Zhang, H. Liu and X. Chen, “IoT-Based Smart Rehabilitation System for Upper Limb Therapy,” *IEEE Access*, vol. 12, pp. 34567–34578, 2024.

[2] R. Kumar and P. Singh, “Visual Cue-Based Reaction Time Training Device for Neuromuscular Rehabilitation,” *Biomedical Signal Processing and Control*, vol. 84, p. 104821, 2023.

[3] S. Lee, J. Park and D. Kim, “Mobile Health Applications for Monitoring and Improving Upper Limb Rehabilitation,” *Journal of Medical Systems*, vol. 47, no. 6, p. 102, 2023.

[4] M. Fernandez, R. Garcia and A. Torres, “Wireless Sensor-Based Smart Therapy Device for Hand Function Recovery,” *Sensors*, vol. 22, no. 15, p. 5678, 2022.

[5] K. Park and H. Kim, “Gamified Rehabilitation Systems to Enhance Cognitive and Motor Training,” *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 30, pp. 1987–1996, 2022.